CHAPTER 6

From Maxillary and Mandibular Final Impression Technique to Set-up of Denture Teeth

1 Initial condition

Pretreatment : Occupation: She is a housewife and mother of three.
Chief complaints : Periodontitis : The patient presented with poor aesthetics and masticatory disturbance caused by severe periodontitis.
Post-treatment : All natural teeth were extracted rendering the patient edentulous.

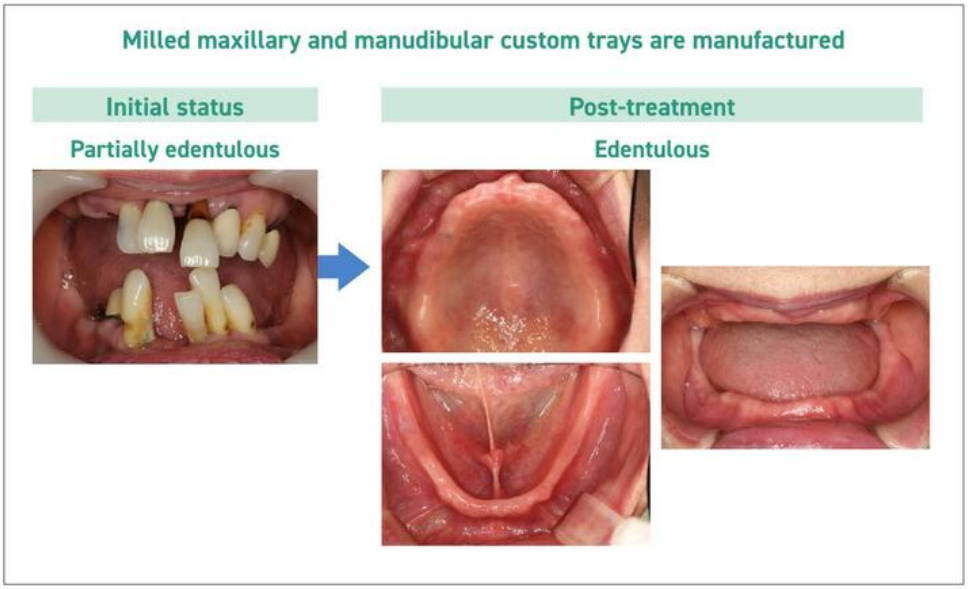


Fig 6-1

2 Try-in of Gnathometer M or CAD equipped custom trays

[1] Checking the occlusal plane of Gnathometer M or Gnathometer CAD

The final impression-taking process begins after the Gnathometer M has been mounted on the casts using the Horizontal Guide. The custom trays are tried in the mouth before impression taking. The patient is asked to bite on the occlusal template to check midline coincidence and parallelism of the horizontal plane with the interpupillary line in the frontal view, and parallelism of the template with Camper's line (naso-auricular plane) (**Figs 6-2,3**).

Gnathoemeter CAD (BP=0, CP=0) is substituted.

This section will describe the maxillary and mandibular precision impressions in terms of the individual digital custom trays with a built-in Gnathometer CAD.

Occlusal plain parallel to the horizontal plain is the most important key to make even pressure toward the ridge mucosa when taking precision impression.



Fig 6-2a Gnathometer M.
Fig 6-2b,c Insert the custom trays with the Gnathometer M and the Gnathometer CAD.



Fig 6-2d Gnathometer CAD.
Fig 6-2e Upper & Lower milled custom trays with seven ideas to enhance suction effectiveness.

Frontal view

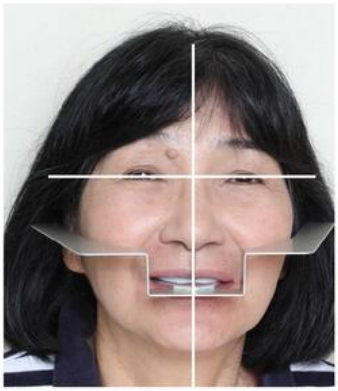


Fig 6-3a The patient is asked to bite on the occlusal template (marketed by J. Morita Corporation; manufactured by Senjo Co., Ltd.) to check the midline and horizontal plane.

Lateral view



Fig 6-3b Check the parallelism of the template with Camper's line (naso-auricular plane).



Fig 6-3c UTS CAD (BP=0, CP=0) is substituted.

[2] Checking errors in intermaxillary relationship

The bite registration taken with the Centric Tray is not perfect. The final occlusal position is established by confirming the patient's OVD after the final impression taking and determining his/her horizontal occlusal position with Gothic arch tracing on the Gnathometer M or CAD. Major errors may occur in the Gnathometer mounting process. These errors should be corrected before proceeding to the final impression. Corrections can be made by placing wax in 14,16,24 and 26 areas (**Fig 6-4**).



Fig 6-4a Milled custom tray.

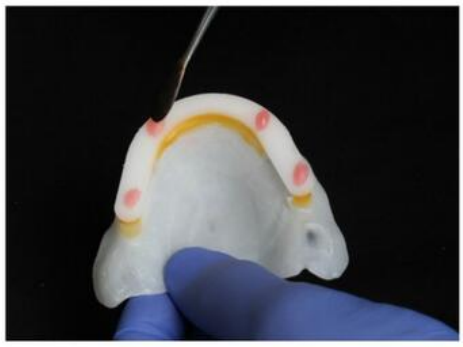


Fig 6-4b Place small amounts of paraffin wax in 14,16,24 and 26 areas and soften them.



Fig 6-4c Insert the mandibular custom tray into the mouth first, next, maxillary custom tray into the mouth and ask the patient to bite.





Fig 6-4d,e Check carefully to see if the maxillary and mandibular plates are in even contact anteroposteriorly and laterally. A slight change in OVD with the addition of paraffin wax will not affect the final impression. The final impression process can be initiated when the operator is certain that the maxillary and mandibular plates are in even contact.

3 Maxillary final functional impression

The maxillary final impression can be successfully taken following border molding with compound in all cases except those with flabby gums and pronounced resorption of the maxillary tuberosity regions. This is because the movable volume of the mucobuccal fold in the maxilla is 1/2 to 1/3 of that in the mandible, which limits the amount of change in the shape of the border of the impression even if the operator over-retracts the cheeks and lips during border molding with softened compound.

In the BPS denture fabrication system, a heavy-body silicone material (Virtual Heavy Body®) is used instead of compound for border moulding with the patient's own functional movements. The final impression is then taken with a high-flow injectable material (Virtual Light Body®). The functional impression obtained with the patient's own movements is far less prone to error, particularly when the dentist is less experienced in complete denture treatment. The final impression procedure of BPS is designed in such a way that almost the same results can be obtained irrespective of the skill level of the operator, which is an important contribution to the simplification of impression procedure.

[1] Maxillary impression

The final impression procedures start with the maxilla. Because of a larger surface area of the denture base in the maxilla than the mandible, the maxillary impression tray tends to be displaced more easily by impression materials. If the tray gets displaced, the occlusal position determined by the operator in the earlier step will not be reproduced accurately. This is the reason why the maxillary final impression is taken first.

The following series of photographs will visually demonstrate the steps of maxillary final impression (**Figs 6-5~22**)

In this part, I explain a precision impression process using the custom tray which is manufactured through digital technology. The result of the impression is the exactly same even if you use the handmade custom tray.

1) Risk factors on Maxillary denture suction

Ridge condition is fair without any inflammation. Small torus is present on the middle palate region. There are no risk factors to affect maxillary complete denture suction.



Fig 6-5 Maxillary ridge condition is very fair.



Fig 6-6 Apply Vaseline around the patient's mouth to prevent the impression material from adhering to the skin.



Fig 6-7 Try-in milled custom trays.



Fig 6-8 Interior surface of the maxillary custom tray and tray adhesive.



Fig 6-9 Apply an adhesive for silicone.



Fig 6-10 Place Virtual Heavy Body® for border molding.

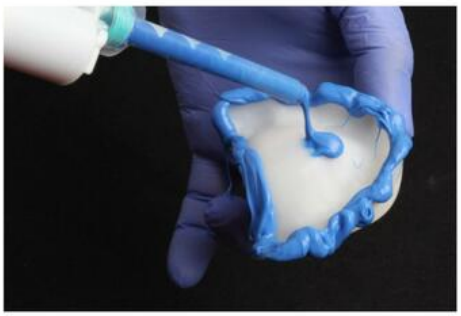


Fig 6-11 The border molding material should cover the entire tray border and apply it as a stopper on the hard palate region.



Fig 6-12 The use of a cheek retractor by the assistant prevents the impression material from touching the lip when the tray is carried into the mouth. Insert the mandibular custom tray first, next maxillary tray. Ask the patient to close the mouth.



Fig 6-13a,b Pull the buccal mucosa in the left premolar area inferiorly and postero-inferiorly, and repeat the same on the right side.
Pull the upper labial frenum downward.



Fig 6-14a,b Ask the patient to move her jaw, side to side to avoid the condyloid process.



Fig 6-15 Excessive impression material is removed to prevent gagging.

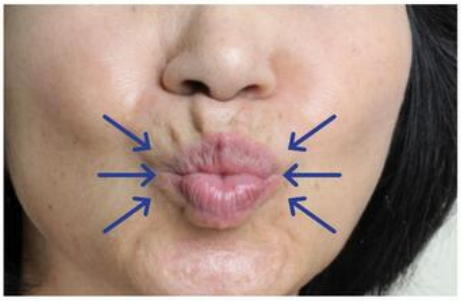


Fig 6-16 Ask the patient to make two functional movements. The first movement is to purse the lips by saying "woo".

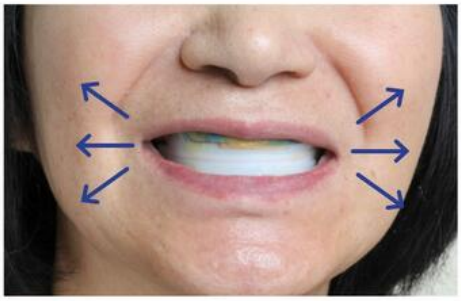


Fig 6-17 The second movement is to retract the corners of the mouth by saying "eee".

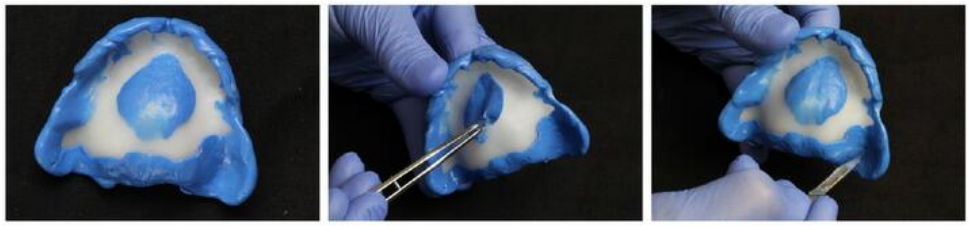


Fig 6-18a~c Remove the tray from the mouth. Trim excess material on the hard palate and immobile ridge mucosa. with a dental carver. This completes the border molding procedure.

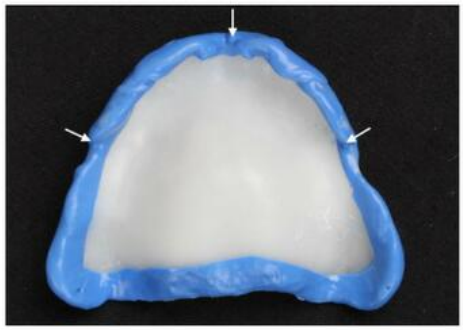


Fig 6-19 Cut the impression material off on the buccal and labial frenum region with the dental scalpel.





Fig 6-20 Syringe the high-flow Virtual Light Body® material evenly onto the tray and insert the loaded tray into the mouth.

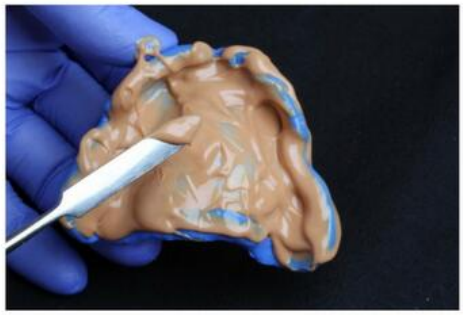


Fig 6-21 Insert the mandibular custom tray next. Take the final impression in the same manner as the preliminary impression.

Completion of the maxillary final impression

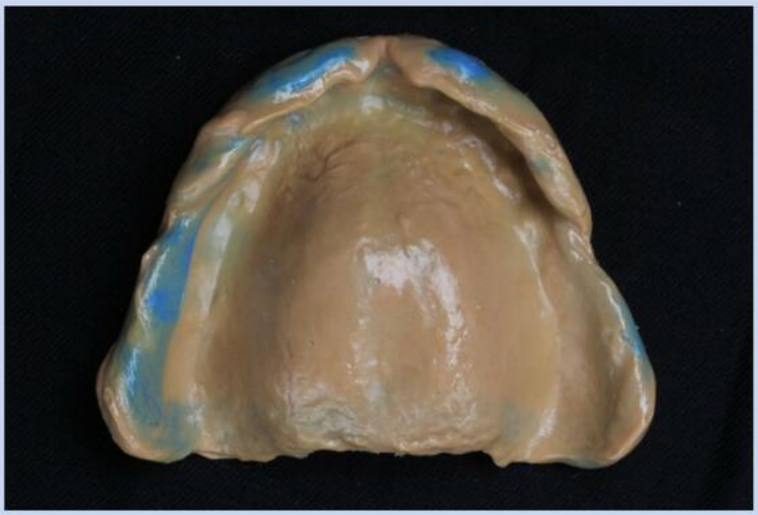


Fig 6-22

4 The suction-effective mandibular final impression -With a focus on the closed-mouth functional impression-

[1] Key points to consider for the mandibular suction-effective final impression

The most important objective of the mandibular final impression intended to achieve suction is to seal the entire denture borders by taking advantage of the characteristics of the oral mucosa. It is most desirable that all borders of the custom tray are tightly sealed by the mucosa while permitting the patient's unique oral mucosal movements. We should change our way of thinking from the conventional concept of denture base extension to the sealing of the entire denture borders, and concentrate our efforts on creating the BTC Point and achieving intimate contact of the interior surface of the denture base with the mucosa in the retromolar pad region, the most difficult region to obtain a good seal.

1) The reason to recommend impression procedure at closed position

It is important for mandibular suction-effective impression to take both preliminary and final impression at closed position (**Fig 6-23**). The condition that negative pressure inside the denture base generates is at the moment the patient bites in from mandibular rest position.

Saliva under the denture base is discharged at that biting moment generating negative pressure by sealing the denture border all around. **Fig 6-23** Upper illustrates relationship of the retromolar pad and the denture border when the denture is fabricated at closed mouth condition. The retromolar pad is stretched posteriorly changing the shape thinner when the mouth is opened and it contracts and gets shortened and rounded when the mouth is closed.

Impression is taken at closed position covering the retromolar pad with the denture base completely. Deformation of the retromolar pad upon opening the mouth is controlled by the denture base, since posterior part of the retromolar pad consists of soft glandular tissue. Lack of direct muscle attachment to the area is another reason that the denture base covering retromolar pad completely could prevent deformation of the retromolar pad.

Fig 6-23 Lower illustrates fabrication of the denture at opened mouth condition. If the impression is taken at opened mouth condition with the retromolar pad stretched and thinned, it creates space underneath the denture base once the retromolar pad contracts when the mouth is closed breaking the border seal.

The principle for suction-effect of maxillary and mandibular denture is complete border seal of the entire denture borders. The maxillary denture would fall down and the mandibular denture would lift up if there is even one spot of air leak. Due to the reasons above, it is important to take preliminary and final impressions in a mouth closed position in order to achieve mandibular suction- effective complete denture.

2) Importance of impression materials

The denture sinks in slightly when the teeth come into occlusion from the mandibular rest position. A negative pressure is generated as a result. This is the very moment mandibular denture suction occurs. It is therefore important to take an impression of the shape of the denture border that is in harmony with the patient's functional 5 movements when the teeth are in occlusion. The straightforward case is shown. The impression material used in the final impression step of BPS is a highly biocompatible silicone material (Virtual Light Body®). The author regarded Ivo-Seal® (Ivoclar Vivadent) as a material of choice for achieving suction. As it is no longer commercially available, materials similar to Ivo-Seal® are also recommendable, such as Soft-Liner® for tissue conditioning (GC CORPORATION) and Tissue Conditioner® (Shofu Inc.) mixed at the powder to water ratio of 10:7 (**Fig 6-24**). A disadvantage of materials of tissue conditioner type with high alcohol content is that the alcohol content evaporates quickly, causing the impression to deform. This necessitates the earliest possible pouring of plaster to minimize deformation.

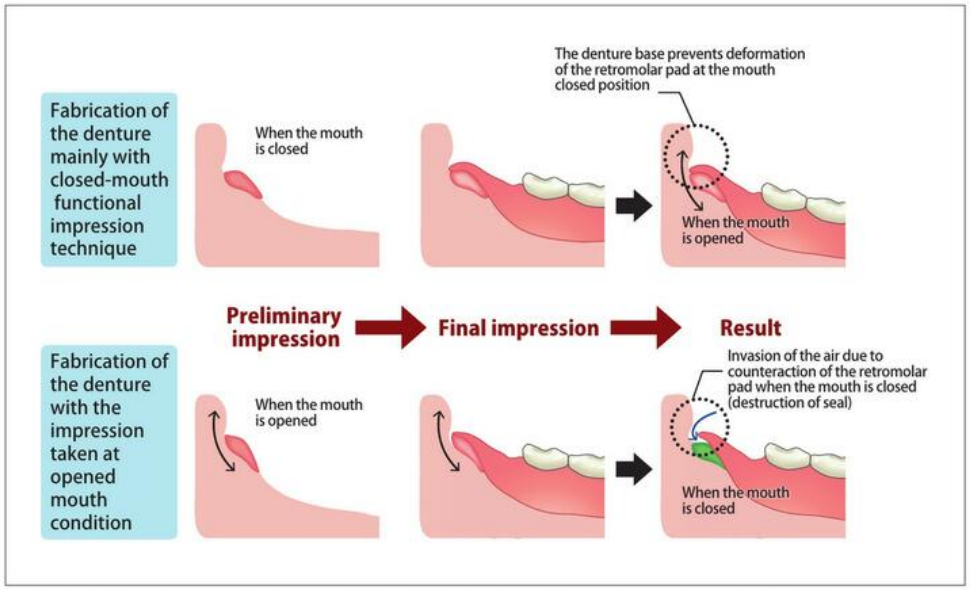


Fig 6-23 The illustrations when the denture is fabricated with closed-mouth impression technique (above) and when the denture is fabricated with opened-mouth impression technique (below).

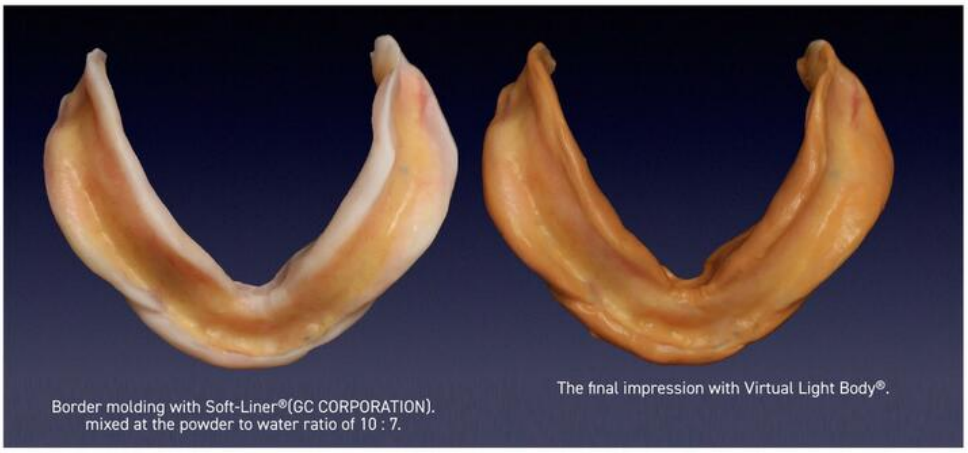


Fig 6-24

5 Precision impression technique achieving mandibular denture suction

[1] Practical mandibular final impression procedure for the easy-to-achieve mandibular suction case

This section will describe the impression methods of an easy-to-achieve mandibular suction case with a favorable ridge condition and difficult-to-achieve mandibular suction case with inhibitory factors of deformable retromolar pads, insufficient sublingual spongy tissue, and tongue retraction.

Easy case for mandibular denture suction. Female : 68years (Fig 6-25)

Examination items for manudibular complete denture — Easy case

Suction inhibiting factor	Right good	Right fair	Right poor	Left good	Left fair	Left poor
1. Ridge form	✓			✓		
2. Spongy tissue in Sublingual fold region	✓			✓		
3. Space for denture extension in retromylohyoid fossa region	✓			✓		
4. Pear-like shaped retromolar pad	✓			✓		

Retromolar pad (one risk is ✓s Fair, ✓s Poor, if more than two risks are marked

	Right			Left		
1. The presence of the fibrous tissue in the anterior half	(Yes	✓ slightly	No)	(Yes	✓ slightly	No)
2. Size	(Large	✓ Moderate	Small)	(Large	✓ Moderate	Small)
3. Tilt angle	(Shallow	✓ Moderate	Steep)	(Shallow	✓ Moderate	Steep)
4. Change during open and close movements	(Small	✓ Moderate	Large)	(Small	✓ Moderate	Large)

5. Tongue retraction upon opening mouth	☑ Normal(less than 2cm)	☐ Further retraction (2~4cm)	☐ Even further retration (more than 4cm)
6. Jaw Relationship	☑ Class I	☐ Class II	☐ Class III
7. Mandibular jaw position	☑ Guided position and habitual occlusal position coincide or within 2mm of discrepancy	☐ More than 2mm of descrepancy	☐ More than 2mm of discrepancy and unstable tapping point
8. Temporomandibular function	☑ Normal	☐ Disfunction	☐ Severe disfunction with pain

Special Note:
Wearing time of denture,(Maxillary) Flabby tissue, Tori(Mandibular) Flaby tissue, Genial tubercle shallow labial vestibule in mandibular anterior region, etc → N / A

Fig 6-25a All itmes are favorable.

(1) Ridge form



Fig 6-25b Favorable mandibular ridge form.

(2) Subgingival tissue



Fig 6-25c Large amount of sublingual spongy tissue.

(3) Retromylohyoid fossa



Fig 6-25d

(4) Tongue position



Fig 6-25e Class1 Normal tongue position.

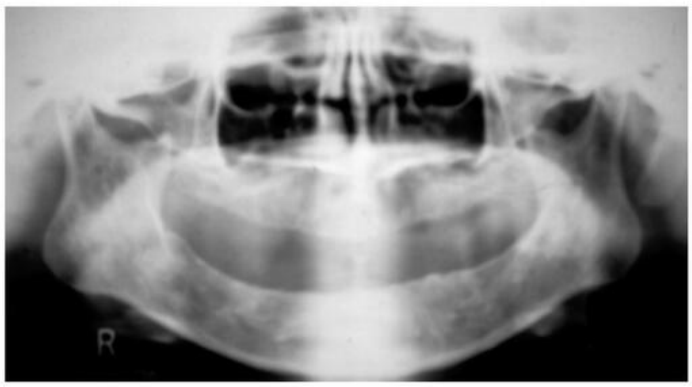


Fig 6-25f

1) Mandibular final impression procedure

Traditionally, border molding has been carried out with compound in sections recording one part of the border at a time. The functional impression technique of BPS is characterised by border molding of all parts of the border simultaneously (**Figs 6-26~28**).

A functional impression is a procedure whereby the border of the custom tray is moulded by the patient's own functional movements and not by the operator. Therefore, the procedure is not affected by the operator's skills. Instead, the outcome of the final impression depends on how correctly the preliminary impression and custom tray fabrication procedures have been carried out.

2) Importance of 5 basic movements for good ridges

For patients with good ridges without inhibitory factors affecting mandibular denture suction, the most important step of the final impression procedure in the mandible (**Fig 6-29**) is the five basic movements the patients are asked to perform as explained on P.15, since the functional impression is complete with these movements in this group of patients.

1) Purse the lips.
2) Say "eee".
3) Lick the upper lip with the tongue.
4) Push the back of the anterior part of the maxillary and mandibular impression trays with the tongue.
5) Swallow.

Generally speaking, a patient with a favorable ridge form has a good range of motion of the lips and cheeks and a high activity of the oral mucosa, making it possible to take a sharp impression of the border areas with functional movements.

The trays are inserted into the mouth and the patient is asked to bite with the trays into place. The retromolar pads are stabilized by the impression material in a well-fitting custom tray, though extremely susceptible to deformation. The shape of the retromolar pads is thus captured in the impression without deformation.

The first two movements the patient is asked to perform are to say "woo" and "eee". These movements shape the labial and buccal borders and provide clearance for buccal frenum movement. The next two movements, moving the tongue from side to side and pushing the anterior lingual area of the tray with the tongue, strongly activate the mylohyoid muscle, which shapes the floor of the mouth. The last movement involving the entire mouth is swallowing. Swallowing causes the buccal mucosa above the retromolar pad to be pulled strongly inward, creating the BTC Point. The exterior seal of the denture border in the retromolar pad region is complete with the formation of the BTC Point. In addition, the mandibular labial border is moulded as the mentalis muscle contracts during swallowing.

Step 1: Border molding



Fig 6-26 Apply Virtual Monophase®, less viscous than Virtual Heavy Body®, to the retromolar pads to minimize their deformation.



Fig 6-27 Apply Virtual Heavy Body® to all denture borders except in the retromolar pad region.



Fig 6-28 The mandibular custom tray loaded with two impression materials of different viscosities,Virtual Monophase® and Virtual Heavy Body®.

Step 2 : The patient is instructed to make 5 basic movements





Fig 6-29a 1) Purse the lips (say "woo") to capture the movement of the lips and buccal mucosa.
Fig 6-29b 2) Retract the corners of the mouth (say "eee"). This also captures the movement of the lips and buccal mucosa.





Fig 6-29c,d 3) Move the tongue from side to side to record tongue movement.





Fig 6-29e 4) Push the back of the tray with the tongue in closed-mouth position, to imprint the floor of the mouth under tension, which reflects a highly contracted state of the mylohyoid muscle.
Fig 6-29f 5) Swallow to capture an activated state of the mentalis muscle, and the overall movement of the oral cavity.

Step 3 : The completed trimming of excess border molding material





Fig 6-30a,b To secure a peripheral seal of the tray, remove excess heavy-body impression material from the interior surface of the custom tray with a scalpel. This maneuver will further enhance the peripheral seal of the final impression. The alveolar mucosa in the border area is highly mobile and compressible. The compression of the alveolar mucosa by the impression material will further enhance the border seal.

Step 4 : the patient is asked again to perform the five basic movements



Fig 6-31 Load the tray with Virtual Light Body® for final impression, and insert the tray into the mouth, and the patient is asked to make the five basic movement again.



Fig 6-32 The final impression completed based to the mechanism of mandibular denture suction.

[2] Practical mandibular final impression procedure for the difficult-to-achieve mandibular suction case

Difficult case for mandibular denture suction. Female : 57 years (Figs 6-33,34)

Examination items for manudibular complete denture — **Difficult case**

Suction inhibiting factor	Right good	Right fair	Right poor	Left good	Left fair	Left poor
1. Ridge form	✓			✓		
2. Spongy tissue in Sublingual fold region			✓			✓
3. Space for denture extension in retromylohyoid fossa region	✓			✓		
4. Pear-like shaped retromolar pad			✓			✓

Retromolar pad (one risk is ✓as Fair, ✓as Poor, if more than two risks are marked

	Right			Left		
1. The presence of the fibrous tissue in the anterior half	(Yes	slightly	No)	(Yes	slightly	No)
2. Size	(Large	Moderate	Small)	(Large	Moderate	Small)
3. Tilt angle	(Shallow	Moderate	Steep)	(Shallow	Moderate	Steep)
4. Change during open and close movements	(Small	Moderate	Large)	(Small	Moderate	Large)

5. Tongue retraction upon opening mouth	☐ Normal(less than 2cm)	☑ Further retraction (2~4cm)	☐ Even further retration (more than 4cm)
6. Jaw Relationship	☑ Class I	☐ Class II	☐ Class III
7. Mandibular jaw position	☑ Guided position and habitual occlusal position coincide or within 2mm of discrepancy	☐ More than 2mm of descrepancy	☐ More than 2mm of discrepancy and unstable tapping point
8. Temporomandibular function	☑ Normal	☐ Disfunction	☐ Severe disfunction with pain

Special Note:
Wearing time of denture,(Maxillary) Flabby tissue, Tori(Mandibular) Flabby tissue, Genial tubercle shallow labial vestibule in mandibular anterior region, etc → N / A

Fig 6-33 Examination sheet for suction-effect of the mandibular complete denture.

According to the outcome of her intra-oral examination, it is obvious that she has three inhibitory factors affecting mandibular denture suction.

Inhibitory factors

Risk factors on Mandibular denture suction

· Poor retromolar pads in shape

· Insufficient amount of sublingual spongy tissue

· Tongue retraction with 3cm from the anterior ridge in mouth opening.

Solution

To solve these problems, two optional impression techniques performing before final precision impression are prerequisite,

1. Optional impression technique to enhance the denture base fit using mono-phase PVS at the retromolar pad area.
2. Optional sublingual sealing technique using silicone putty.

(1) Ridge form



Fig 6-34a Favorable ridge condition to make dentures stable.

(2) Sublingual spongy tissue



Fig 6-34b Insufficient sublingual spongy tissue to affect the lingual denture border sealing.

(3) Retromylohyoid fossa

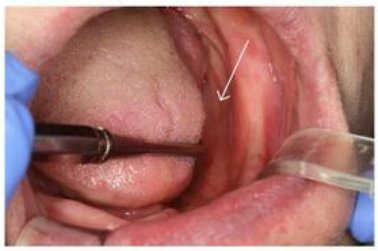


Fig 6-34c Sufficient space at the retromylohyoid fossa where the denture border can be extended over the mylohyoid line in order to complete the compensatory closure.

(4) Retromolar pads



Fig 6-34d Lack of fibrous tissue at the medial part of the retromolar pads.



Fig 6-34e Mild curvature of the inclination angle of the retromolar pads.

(5) Tongue retraction

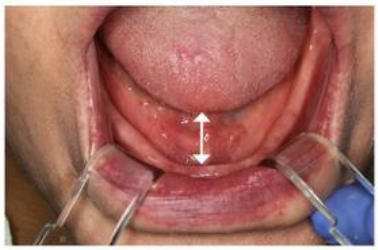


Fig 6-34f The tongue retracts in the mouth closed within 3cm from the anterior ridge.

1) Optional impression technique to improve the denture base fit at the retromolar pad area (Figs 6-35,36)

This technique should be carried out only in a case with poor retromolar pads that have been identified because of the intra-oral examination.

The author highly recommends the use of Virtual Monophase® impression material over Virtual Heavy Body®, as Monophase is less viscous and minimizes the deformation of the retromolar pads.

Procedure

While paying careful attention not to distort and overextend the retromolar pads, a small amount of Monophase impression material is applied only on the interior side of the tray at the retromolar pads area, and the patient is asked to close the mouth and swallow. The impression material captures the boundary between the pads and buccal mucosa so that the natural shape of the retromolar pads is recorded when the mouth is closed.

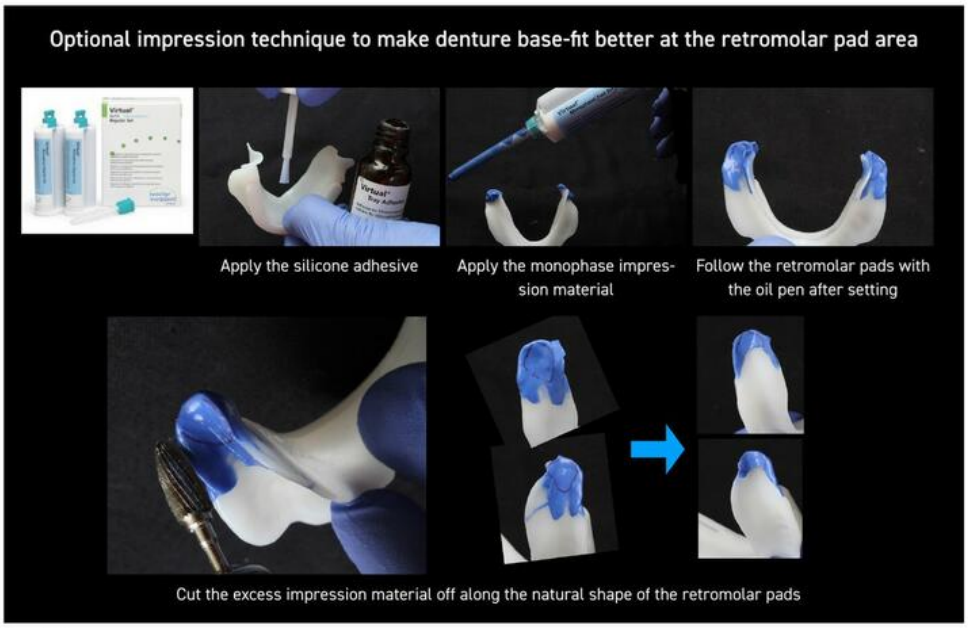


Fig 6-35

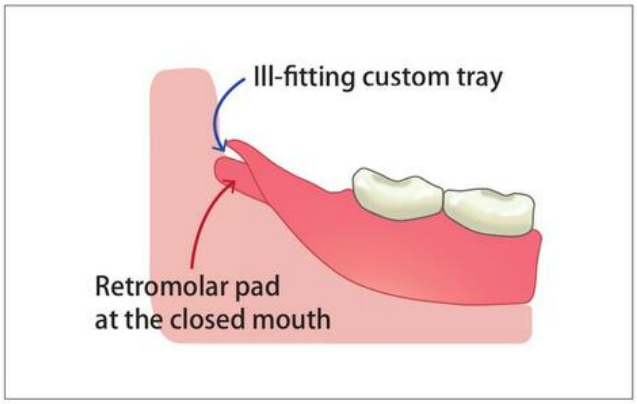


Fig 6-36

2) The optional sublingual sealing technique in a case with the tongue retraction

Insufficient sublingual spongy tissue and tongue retraction while opening the mouth are anatomical suction inhibiting factors of mandibular complete denture. These inhibiting factors are challenging to manage because an air leaking space is easily created around the sublingual denture border (**Fig 6-37**). This space can often be managed by utilizing an optional seal-effective impression technique with silicone putty before the basic five functional mouth movements are performed (**Fig 6-38**). Be assured that this optional technique is not needed in all cases.

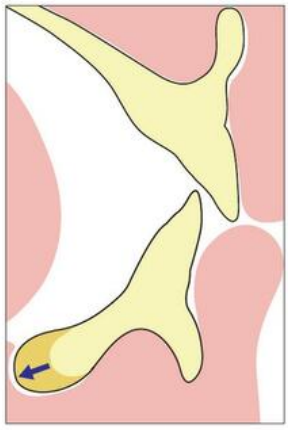


Fig 6-37 Air leaking space is created by the tongue retraction. The lingual space is filled with the putty.

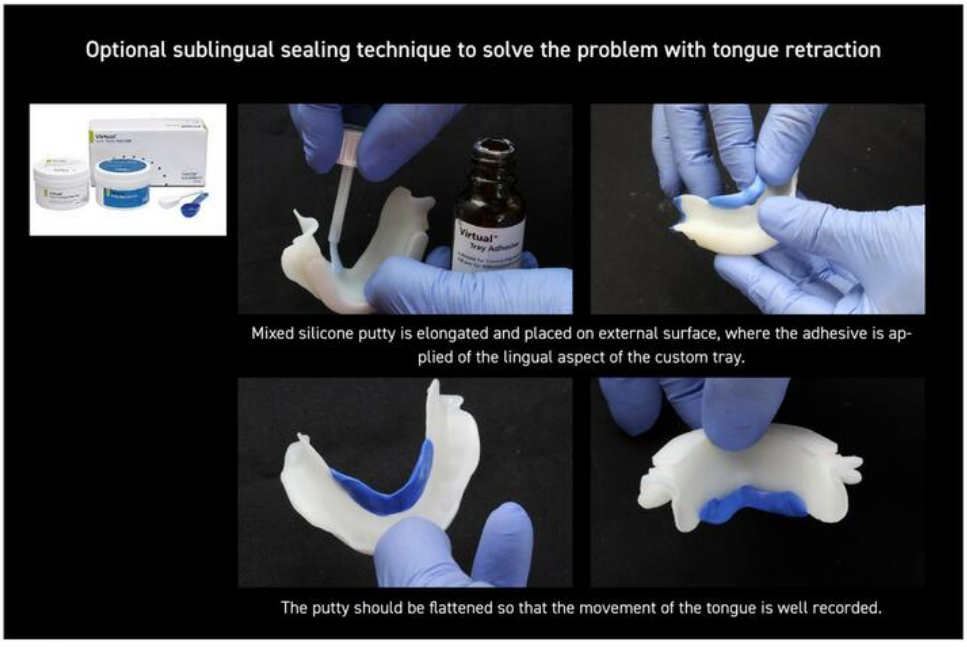


Fig 6-38

1 Optional lingual sealing technique procedure

In severe cases with inhibiting sublingual factors, impressions are taken by having the patient repeat the following sequence of movements several times (**Fig 6-39**):

(1) Swallowing is a comprehensive oral movement that involves the entire mouth.

(2) The lingual area of the mandibular impression tray is painted with VPS adhesive, then a small roll of silicone putty is applied to the tray. After inserting the tray, the patient swallows. Then the patient opens the mouth carefully so that the custom tray will not lift. The operator proceeds to hold the tray firmly down for 10 seconds. A few drops of water are placed in the mouth with a pipette. The patient swallows the water, then opens the mouth again. The operator holds the tray for another 10 seconds; then, the patient swallows again. This cycle is repeated several times until the impression material begins to set. The operator must pay close attention to secure the tray in place to prevent it from lifting off the ridge throughout this process. If the tray lifts even once, it is recommended to discard the impression and start over again. These two movements aid in achieving a good border seal with challenging cases (**Fig 6-40**)

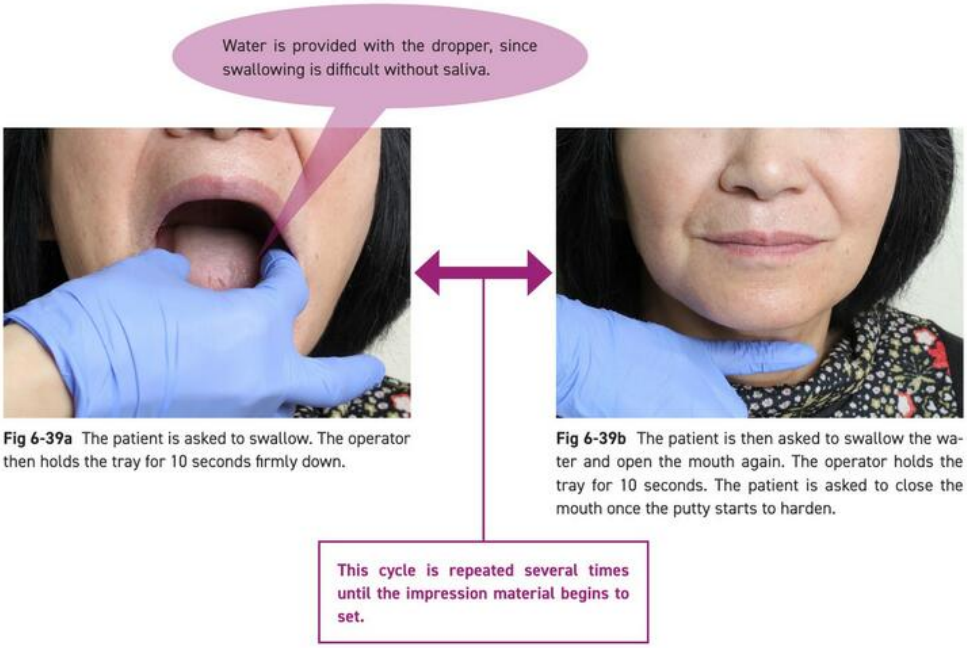


Fig 6-39a The patient is asked to swallow. The operator then holds the tray for 10 seconds firmly down.

Fig 6-39b The patient is then asked to swallow the water and open the mouth again. The operator holds the tray for 10 seconds. The patient is asked to close the mouth once the putty starts to harden.

We are ready to approach to initiate final suction-effective precision impression!

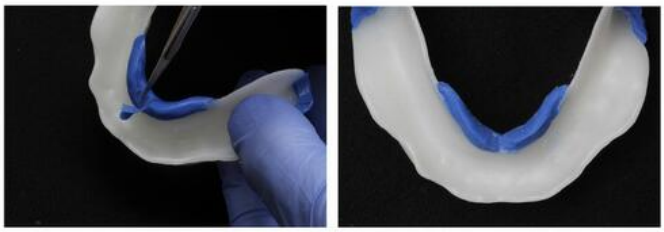


Fig 6-40a And the impression material on the lingual frenum is cut off to avoid overpressure by the second impression material.

Fig 6-40b The impression material on the interior side of the custom tray should be removed to make space for second impression material.

[3] Mandibular final impression procedure following two optional impression techniques

After finishing two optional impression techniques around the retromolar pad and sublingual region, the patient is instructed to make 5 basic movements (**Figs 6-41~43**).

Preparation for mandibular final impression

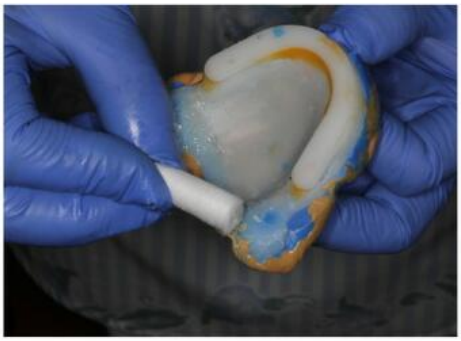


Fig 6-41a Vaseline is applied not to gain the PVS in mandibular impression taking.



Fig 6-41b The mandibular final impression procedure starts with the placement of the maxillary final impression back into the mouth.

1) Basic Impression Steps

First, the tray adaptation is verified with GC Fit Checker or other light- body VPS material. Painful pressure points are ground out, and overextended tray borders are relieved. An ill-fitting tray may be relined with a commercially available relining material to improve its adaptation.

Step 1 : Mandibular border moulding with Virtual Heavy-body



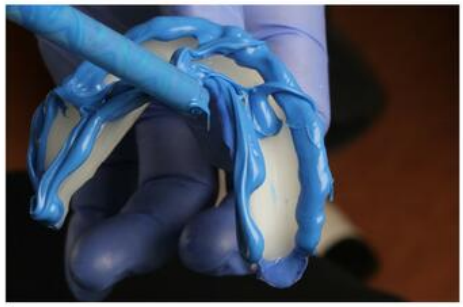


Fig 6-42a,b Apply Virtual Heavy Body® to all denture borders except in the retromolar pad region.

Step 2：The patient is instructed to make 5 basic movements for border molding



Fig 6-43a (1) Purse the lips (say "woo") to capture the movement of the lips and buccal mucosa.



Fig 6-43b (2) Retract the corners of the mouth (say "eee"). This also captures the movement of the lips and buccal mucosa.



Fig 6-43c (3) Move the tongue from side to side to record tongue movement.

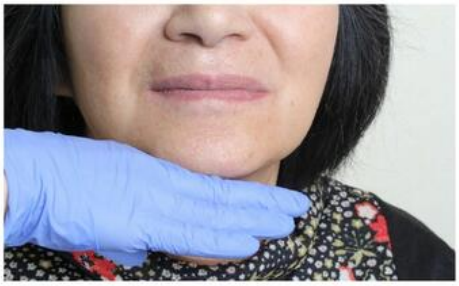


Fig 6-43d (4) Push the back of the tray with the tongue in closed-mouth position, to imprint the floor of the mouth under tension, which reflects a highly contracted state of the mylohyoid muscle.



Fig 6-43e (5) Swallow to capture an activated state of the mentalis muscle, and the overall movement of the oral cavity.

Step 3：Removing the excess impression material

To secure a peripheral seal of the tray, remove excess heavy-body impression material from the interior surface of the custom tray with a scalpel. The alveolar mucosa in the border area is highly mobile and compressible (**Fig 6-44**). The compression of the alveolar mucosa by the impression material will further enhance the border seal.

The immobile ridge mucosa, on the other hand, will not be easily compressed by the impression material. Because of the difference in compressibility, removing the border molding material from the internal surface of the custom tray allows space to record the immobile ridge mucosa and reinforces the border seal.



Fig 6-44a Border molding completed based on 5 basic movements.

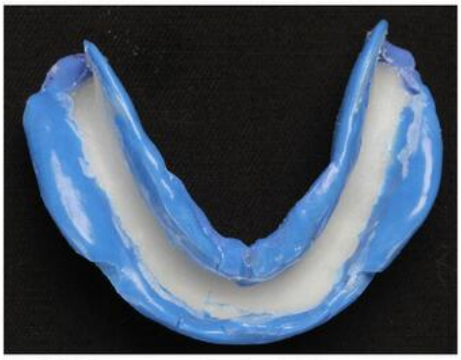


Fig 6-44b To secure a peripheral seal of the tray, remove excess heavy-body impression material from the interior surface of the custom tray with a scalpel.

Load the tray with Virtual Light Body® for final impression while paying carefully attention to spread the impression material out evenly, buccolingual and labial polished surface with a spatulavand insert the tray into the mouth. Finally definitive impression is taken by 5 basic movements.

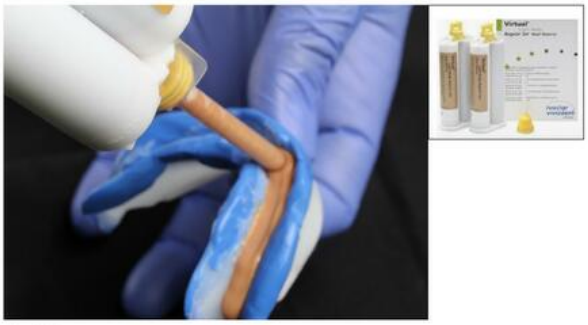


Fig 6-45a



Fig 6-45b

Step 4：The final impression according to perform the 5 basic movements

The final impression that represents the patient's oral function (**Fig 6-46a**) has now been obtained. Let us compare a conventional final mandibular impression with a suction-effective mandibular complete denture whose entire border is sealed by the mobile oral mucosa (**Fig 6-46b**).

Varying thicknesses and lengths of the lingual border play an essential part in achieving suction. Note the impression of the thick border in the sublingual fold region. This border allows full utilization of the spongy tissue in the sublingual fold region to promote suction (**Fig 6-47a**). In contrast, the thickness of the border impression captured in the retromylohyoid fossa region is thin to establish the compensatory closure of the denture base in this region (**Fig 6-47b**). Someya's sinew strings are well delineated. Furthermore, the use of Virtual Monophase®, which is slightly less viscous than Virtual Heavy Body®, permits an accurate recording of the shape of the retromolar pad in a mandibular rest position (**Fig 6-47c**). Attention to these details allows for close-contact-closure between the interior surface of the denture base and the mucosal lining over the retromolar pad. On the labiobuccal side, frenum movements are captured sharply to achieve an interior-exterior double closure (**Fig 6-47d**).

All the requirements for mandibular denture suction have been successfully fulfilled, starting with the preliminary impression using the FCB tray, followed by the design of the custom tray, incorporation of the six ideas for suction into the impression tray, and the final impression taking in the correct occlusal relationship.

Many SEMCD lecture attendees have expressed very similar comments and often share their experiences, such as "Hearing the suction "Pop" sound, made me very happy and I got goosebumps" or, "I cannot wait to apply this technique in my practice to create more patient satisfaction". It is evident that when you apply simply the suction "Pop" sound as the primary key element to evaluate the quality of the edentulous impression of the mandible, you can confidently measure the quality of your impressions and present your advanced skill level to your patients.



Fig 6-46a The final impression is succeeded based to the mechanism of mandibular denture suction.

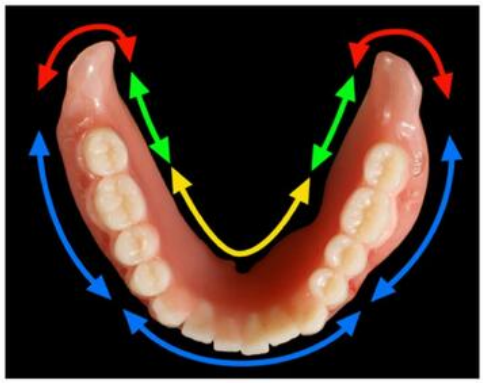


Fig 6-46b The mechanism of mandibular denture suction.

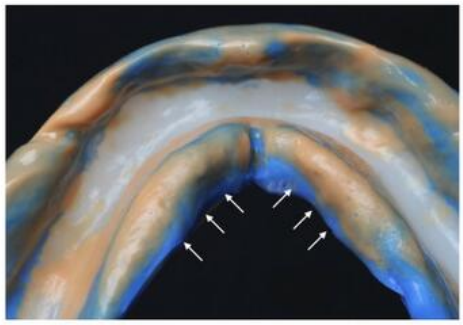


Fig 6-47a Sufficient thickness of the sublingual border.

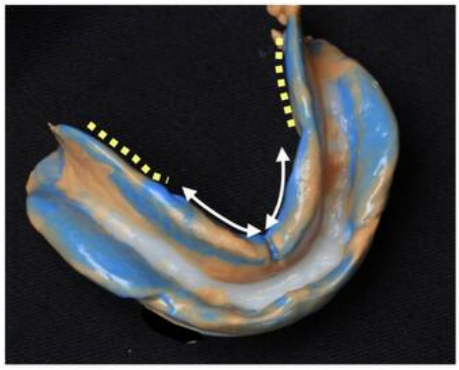


Fig 6-47b Thinness of the retromylohyoid border.

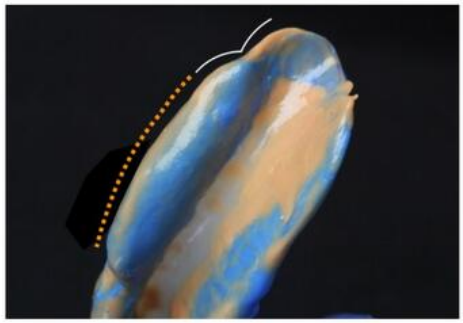


Fig 6-47c Thickness of the buccal shelf border.



Fig 6-47d Lubial and buccal frena are captured sharply.

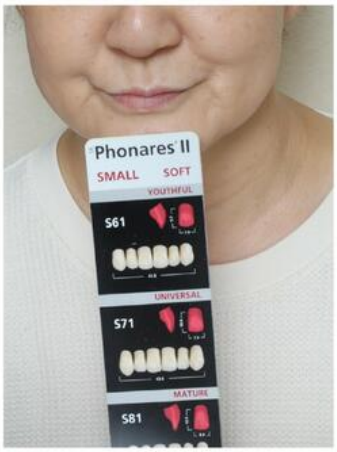
6 Denture tooth selection

[1] Selection of anterior teeth

In the BPS denture fabrication system, the mold and shade of anterior denture teeth are selected using the FormSelector and a shade guide, respectively (**Figs 6-48~51**). The availability of these guides greatly help less experienced dentists, since they no longer have to rely on their subjective judgments. Instead of mechanically selecting prosthetic teeth, it is essential to find out the patient's preferences before making the final decision on the mold and shade, mainly because of an increasing number of patients who want whiter teeth. This section discusses the procedure of denture tooth selection using the SR Phonares teeth.

Fig 6-48 An appropriately sized tooth mold is selected first using the SR Phonares FormSelector (Facial Meter).
Fig 6-49 There are soft and bold tooth forms, which are further divided into three categories (youthful, universal, mature) by age-related severity of occlusal wear.





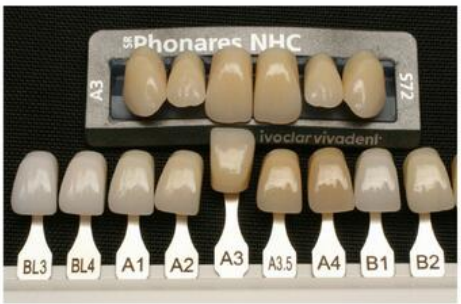


Fig 6-50 The next step is shade selection using the SR Phonares shade guide.



Fig 6-51 The denture teeth finally selected in consultation with the patient.

7 Successful impression or failed impression

[1] How does the operator distinguish the successful suction precision impression and the failed one.

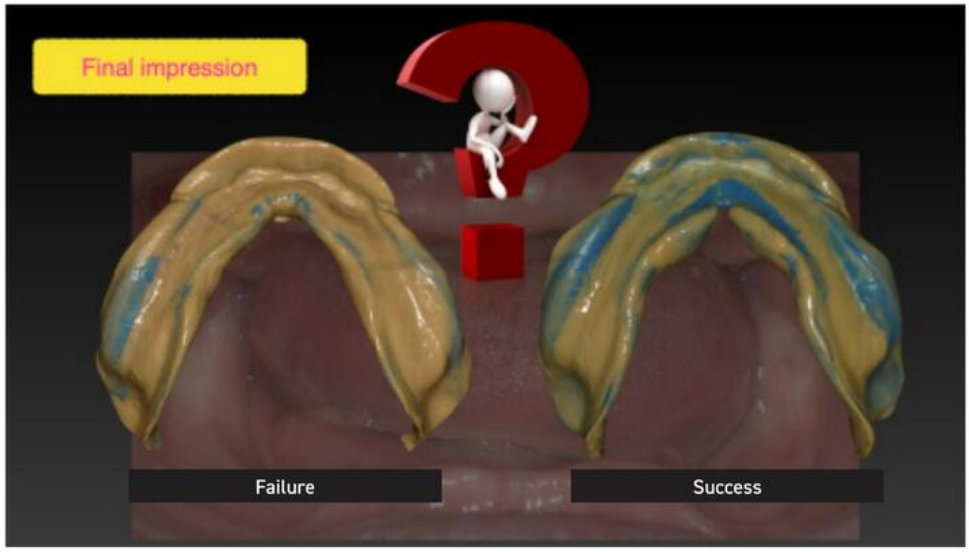


Fig 6-52 Pictures provided by the courtesy of Dr.Masayuki Aizawa.

The impression on your left : Failure (Fig 6-52)
The uneven thickness of light bodied impression material is observed on the heavy bodied impression material due to the tray position error and uneven pressure is applied on the ridge mucosa.

The impression on your right : Success (Fig 6-52)
Even and thin thickness of light bodied impression material is observed on the heavy bodied impression material because of the correct tray position and even pressure in the final precision impression.

This demonstration to say that the correct tray-position and applied even pressure in the final impression phase allow us to take adequate suction impression.

8 Pin-tracing (Gothic arch tracing)

[1] Gothic arch tracing

After determining the OVD, it is common practice to ask the patient to bite to establish a horizontal jaw relationship. However, no textbook states that wherever the patient bites is his/her horizontal jaw position. In this regard, there are discrepancies between what textbooks recommend and what clinicians do in practice. When the horizontal jaw relationship is registered without the aid of a Gothic arch tracing, the operator will not know if the patient's tapping point is stable or if the jaw position is correct.

These errors may become manifest as significant occlusal discrepancies in the final stage of denture fabrication, necessitating re-arrangement of the denture teeth or re-fabrication of the entire denture. If we cut corners, they will come back to haunt us.

Gothic arch tracing is a reliable method whereby mandibular border movements are traced on a registration plate to determine the horizontal jaw relationship. Many dentists do not use this method because of an increased number of visits and additional laboratory costs of fabricating Gothic arch tracers. The most significant advantage of the Gothic arch tracing, however, is that it visualizes the patient's habitual tapping position and apex point. "Visualization" is a feature unique to bite registration with a Gothic arch tracing that no other bite registration technique provides.

It is also noteworthy that the concept of jaw manipulation has changed over the years. In the past, a passive method for gothic arch tracing was recommended in which the operator pushed and guided the chin backward. Presently, the mandible is not manipulated into a retruded position. Instead, a patient-focused active method is preferred in which the patient voluntarily performs mandibular border movements. A reason for this change in the concept of jaw manipulation is the high prevalence of TMJ dysfunction due to structural changes among edentulous patients [35-37].

[2] Gothic arch tracing with either the Gnathometer M or CAD

The patient's provisional OVD is checked after the maxillary and mandibular final impressions have been taken. The final OVD is determined with the use of either Gnathometer M or CAD, which eliminates the need to construct additional bite registration trays for a Gothic arch tracing. Both devices, Gnathometer M and CAD, are highly recommended devices for dentists who recognize the need for Gothic arch tracing but have been avoiding the procedure as it is more cumbersome (**Figs 6-53~56**).

Pin-tracing with Gnathometer M or CAD

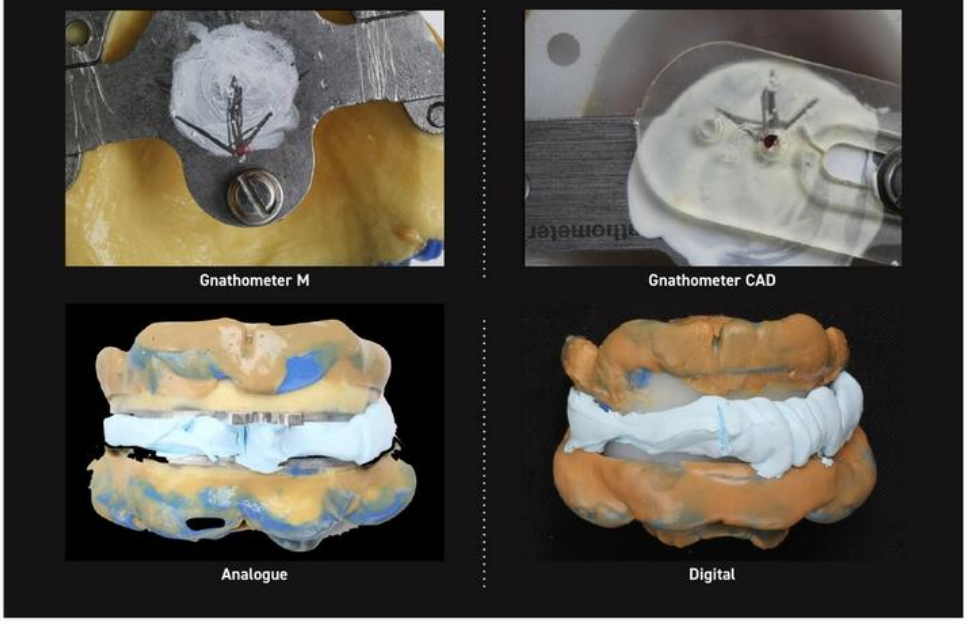
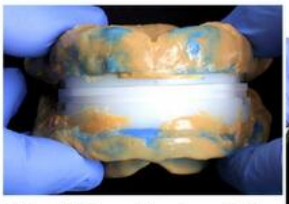


Fig 6-53

Gnathometer CAD



Before Gothic arch tracing with the Gnathometer CAD, OVD is reconfirmed by using phonation and freeway space measurement.





The screw of the Gnathometer CAD is turned with the screw driver and the hight is secured with the sticky wax.

Fig 6-54

Pin tracing with Gnathometer CAD

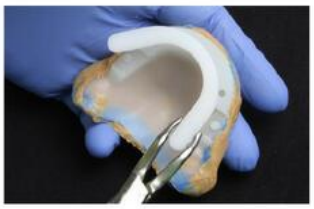



Fig 6-55a,b Remove the bite rim mounts after the maxillary and mandibular final impressions have been taken.



Fig 6-55c Heat the striking metal plate and china marker is painted on it.

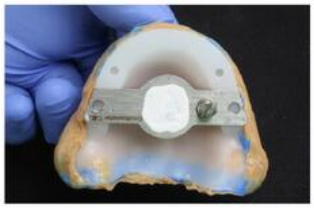
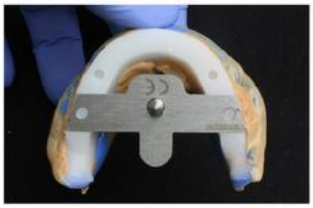



Fig 6-55d, e The mandibular registration plate with a Gothic arch registration stylus is mounted with a pair of long-nose pliers or hand pressure.

Fixation with upper and lower trays



Fig 6-56a Ask the patient to bite in the posterior area and perform tapping movement. Use this posterior tapping position as a starting point for Gothic arch tracing.Next, Ask the patient to slide the mandibular jaw forward, and then back to the starting point, next, to slide the jaw to the left and right and back to the starting point.

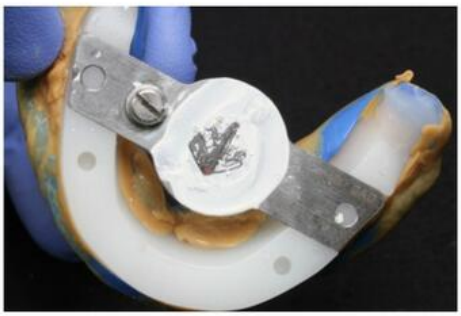


Fig 6-56b Have the patient repeat these movements several times until they can make smooth movements. Place a piece of red articulating paper to record the tapping point.

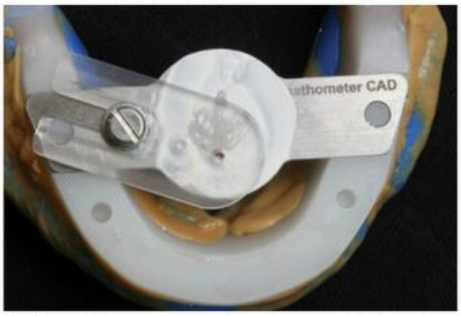


Fig 6-56c The tapping point, if sharp and stable, is used as the final occlusal position instead of the apex point. One of the holes on the transparent plastic fixation plate is matched with the tapping point.



Fig 6-56d Mark the midline and the corner of the mouth, and high lip line.



Fig 6-56e Have the patient close slowly to see if the registration pin goes right into the hole in the fixation plate.

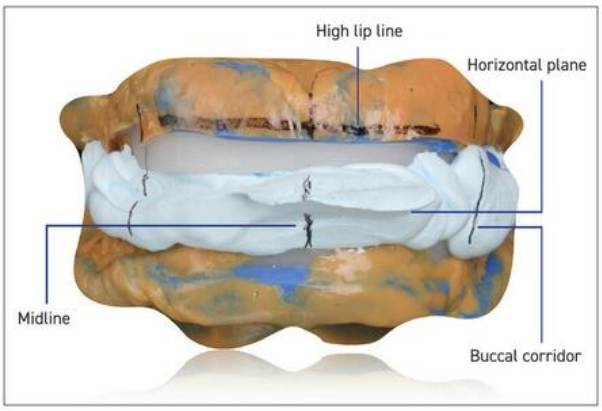


Fig 6-56f Fix the bite in this position with a silicone bite registration material.

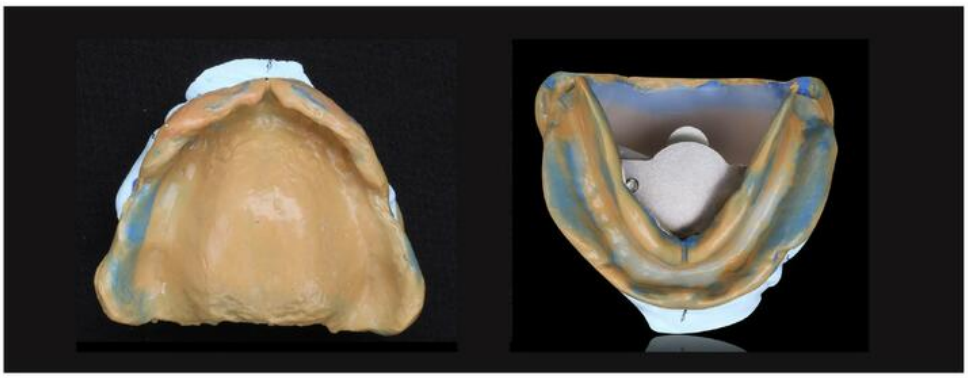


Fig 6-56g Upper and lower final impressions with strong suction effectiveness.

9 Boxing and Articulation (Analog)

[1] Boxing the final impressions (5mm below the borders) and making the casts

The 5mm peripheral border area of the final impression (**Figs 6-57~59**) contains crucial information for achieving suction retention, such as the form and thickness of the denture base. The information must be preserved in the final cast and reproduced with a denture base resin (**Fig 6-60**). Another tip is to use a high-quality mounting stone, such as Elite Arti by Zhermack, with a setting expansion rate of 0.02% after 2 hours. This reduced expansion will minimize the opening of the incisal pin. This section will illustrate the method of boxing aimed at preserving the borders of the impression and replicating them accurately with a denture base resin.

Laboratory procedure for construction of the final casts (Figs 6-57~62)

Boxing

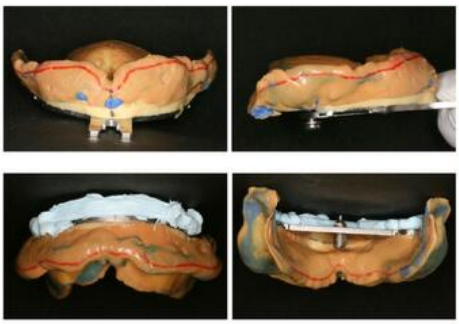


Fig 6-57a~d Draw a block-out line approx. 5 mm from the border of the final impression with an indelible pen.

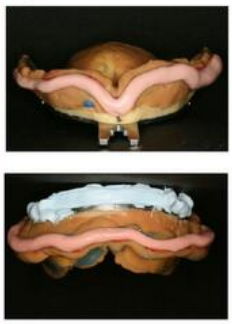


Fig 6-58a,b Block out the borders with a boxing wax.

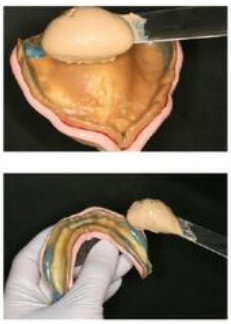


Fig 6-59a,b Pour the impression with plaster having an expansion coefficient that compensates for polymerization shrinkage of the denture base resin. Place the impression face down and wait for the plaster to set.

Mounting the final casts

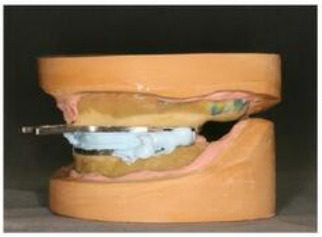


Fig 5-60 Make sure that the Gnathometer M registration fits the maxillary and mandibular casts.



Fig 5-61 The maxillary cast was mounted on the Stratos 300, using UTS. Prepare a split cast for post-polymerization remount or the mandibular cast is mounted on the Stratos 100,200,300 using the horizontal guide.

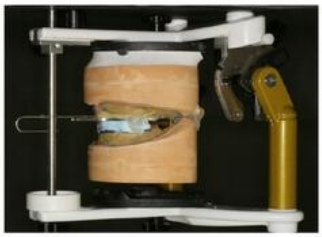


Fig 5-62 Mount the casts on the articulator using the low-expansion mounting stone Elite Arti (Zhermack, feed) with a small setting expansion of 0.02% at 2 hours.

10 Boxing and Articulation (Digital)

[1] Boxing the final impressions (5mm below the borders) and making the casts

Triple scan: To successfully manufacture suction-effective mandibular complete denture and meet the edentulous patient's expectation, we need to digitally scan and record at least 5 mm of the peripheral border area and transfer this information into our digital model and denture, same as with the analog boxing method. This prerequisite laboratory work is foundational to make both analog and digital SEMCD dentures that are suction-effective and make our patients happy.

Maxillary and mandibular impressions are scanned together with a centric bite registration material and appear on the screen as models (**Fig 6-63**).
These models are precisely mounted on a virtual articulator without analog laboratory errors, such as gypsum expansion and mounting issues, while keeping intermaxillary jaw relationships that were carefully established (**Fig 6-64**).

Laboratory procedure for construction of the final casts (Figs 6-63,64)

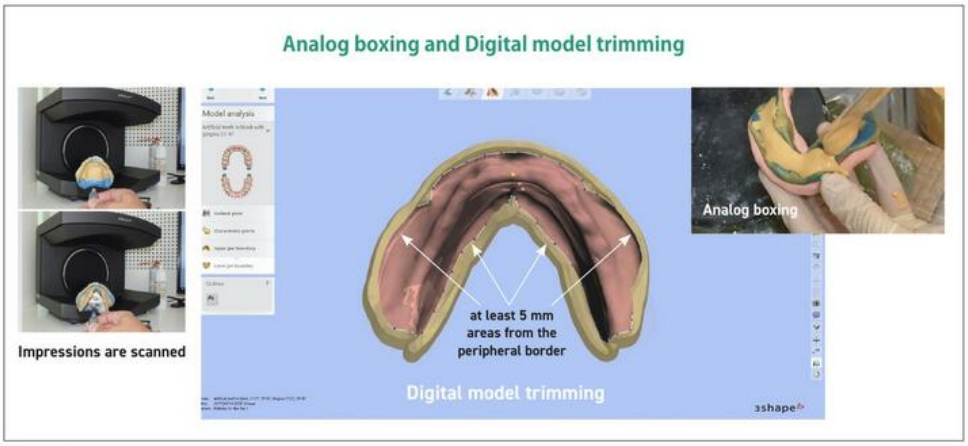


Fig 6-63

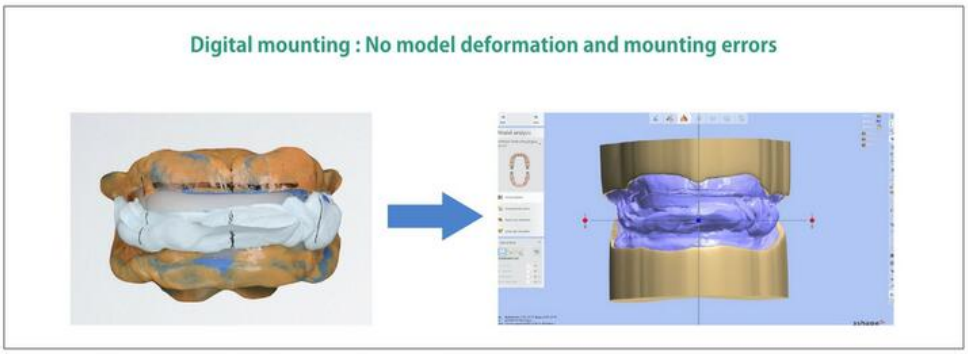


Fig 6-64 These slides are courtesy by Mr.Mattheus Boxhoorn

11 Model analysis on common ground of Analog & Digital

[1] Maxillary model analysis

The model analysis is performed to arrange denture teeth in an appropriate and rational manner. [4, 38] Landmarks for setting denture teeth are identified and drawn on the final casts (**Figs 6-65~67**).

Maxilla

1. Incisive papilla (landmark for anterior tooth set-up.)
2. Fovea palatine (landmark for the midline.)
3. Median palatine raphe (landmark for the midline and posterior border)
4. The first large pair of palatine rugae (landmark for setting canines.)
5. Hamular notches (landmark for the posterior border of the denture.)
6. Maxillary ridge crest lines (landmarks for posterior tooth set-up.)

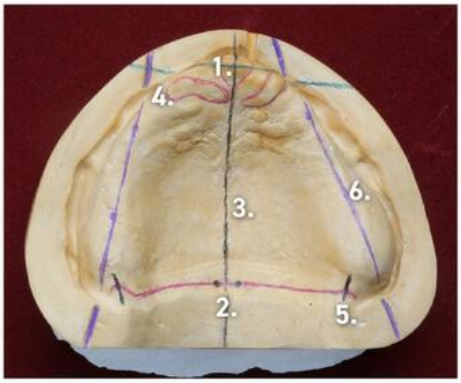


Fig 6-65 Landmarks were marked on the maxillary final cast: midline of the cast, midpoint of the incisive papilla, median palatal raphe, line connecting midpoints of fovea palatini (if inconspicuous, use midpoints of hamular notches).

Fig 6-66 It is pressed on the mandibular ridge to take the impression of the mandible, with unfavorable ridge shape, first. Trace the incisive papilla and the first large pair of palatine rugae. Mark the midpoint of the incisive papilla.

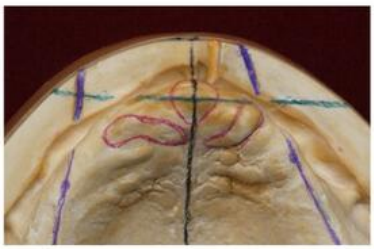




Fig 6-67a~c Draw the maxillary ridge crest line on each side by marking the high points on the crest in the maxillary first premolar area (posterior to the first palatine ruga near the buccal frenum) and first molar area (anterior to the maxillary tuberosity) and connecting the two points.

[2] Model analysis aimed at mandibular denture suction

The SEMCD model analysis aims to seal the entire denture borders with the oral mucosa and defines occlusal loading on the residual alveolar ridges. BPS prioritizes the set-up of denture teeth on the mandible, which tends to be more problematic than the maxilla. Thus, mandibular landmarks determine the occlusal plane for the placement of posterior teeth and are drawn on the mandibular cast (**Figs 6-68~73**).

Mandible

1. Trace the retromolar pads (reference for the posterior denture border).
2. Draw horizontal line at distal 1/3 of the retromolar pads (reference for the occlusal plane).
3. Draw a line bisecting the buccolingual width of the ridge and its extension on each side (reference for setting the posterior teeth).
4. Draw a line bisecting the labiolingual width of the anterior ridge and its extension (landmark for setting the anterior teeth).

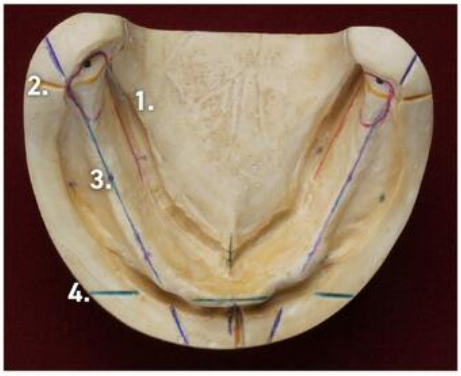


Fig 6-68



Fig 6-69 The mandibular posterior teeth are arranged in the buccolingual center of the ridge or neutral zone where it is easier to achieve the BTC point critical for mandibular denture suction.

Laboratory Procedure ①② Drawing a horizontal line at distal 1/3 of the retromolar pad

Fig 6-70 Draw a horizontal lines at distal 1/3 of the retromolar pad as reference for the occlusal plane, and a line bisecting the buccolingual width of the ridge on each side.



Laboratory Procedure ③〜⑥ Drawing a reference line for the setting of the posterior teeth

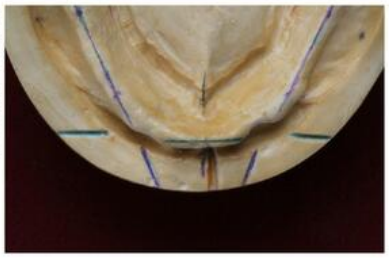
Fig 6-71a Determination of the first molar position. Measure the distance between the lowest points on the buccal and lingual borders.
Fig 6-71b Mark the midpoint of the distance.



Fig 6-72a,b Determination of the first premolar position. Measure the distance between the lowest points on the buccal and lingual borders around the buccal frenum. Mark the midpoint of the distance on the cast. When the midpoint is outside the ridge, mark the buccal edge of the ridge. Connect this point and the previously drawn midpoint for the first molar with a line. Use this line as reference for setting the mandibular posterior teeth.



Fig 6-73 Draw the centerline of the mandibular anterior ridge as reference for setting the mandibular anterior teeth. The intersection of this line (Green) and the mandibular posterior line (blue) helps determine the axial inclination of the mandibular canine.



12 Occlusal scheme of BPS

What occlusal scheme should be selected for edentulous patients? With the recent trend towards Evidence-Based-Dentistry (EBD), there has been an increasing number of reports published from randomized clinical studies on occlusal schemes for edentulous patients.

One of these reports has shown that no statistically significant difference was found in patient satisfaction between the traditionally recommended balanced occlusion and canine guidance[60], further escalating the controversy over what occlusal scheme should be applied to complete denture treatment.

However, the report has neither denied the balanced occlusion concept nor stated that canine guidance is more advantageous. In the author's opinion, this question is likely to remain inconclusive for quite some time until new scientific evidence emerges. Under these circumstances, it may be safe, though conservative, to continue to use the scheme of balanced occlusion that has been accepted by practicing clinicians for so many years without significant issues. A lingualized form of balanced occlusion in which only the maxillary lingual cusps are brought into contact[61] has been recommended, particularly for patients with severely resorbed ridges (**Figs 6-74, 75**). This occlusal scheme is commonly called a lingualized occlusion.

BPS follows the scheme of semi-balanced occlusion for one-tooth-to-two-teeth relationship. To achieve this occlusal scheme, denture teeth are adjusted so that the premolars on the working side and the molars on the balancing side glide simultaneously during lateral excursions. For a lingualized occlusion with a one-tooth-to-one-tooth relationship, two premolars and two molars on the working side and the same four teeth on the balancing side contact and glide at the same time during lateral excursions.

During protrusive movement, adjustments are made so that several anterior teeth and several posterior teeth contact simultaneously for anteroposterior balance.



Fig 6-74a,b Occlusal contacts (green) on the SR Phonares prosthetic teeth in right lateral excursion in lingualized occlusion.

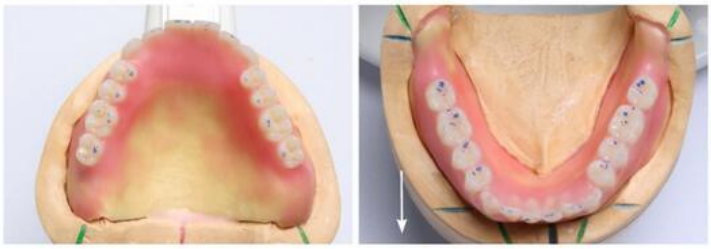


Fig 6-75a,b Contacts (blue) on the SR Phonares prosthetic teeth in protrusion in lingualized occlusion.

13 Setting denture teeth (lingualized occlusion) (Analog)

The following pages describe the set-up process using the nano-hybrid composite SR Phonares teeth that are indicated for not only complete dentures but implant-supported overdentures.

[1] Setting the maxillary anterior teeth (set-up sequence: 11,21→ 13,23 → 12,22)

Laboratory procedure (Figs 6-76~84)

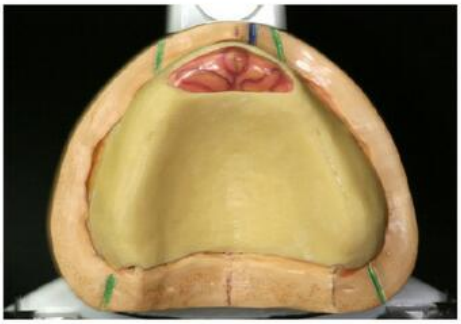


Fig 6-76 Hollow out the anterior part of the baseplate to expose the incisive papilla and palatine rugae landmarks for anterior tooth set-up.

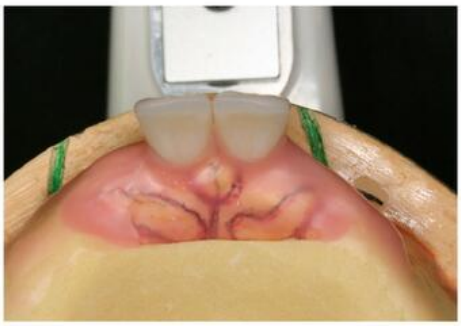


Fig 6-77 The cervical areas of the maxillary central incisors are positioned in line with the center of the incisive papilla anteroposteriorly.



Fig 6-78 The incisal edges are placed at half the distance between the maxillary and mandibular mucobuccal folds measured at the bases of the upper and lower labial frena plus a vertical overlap (1.5 to 2.0 mm). This patient has 40 mm of distance between the maxillary and mandibular mucobuccal folds. The incisal edges are thus placed at 22 mm.

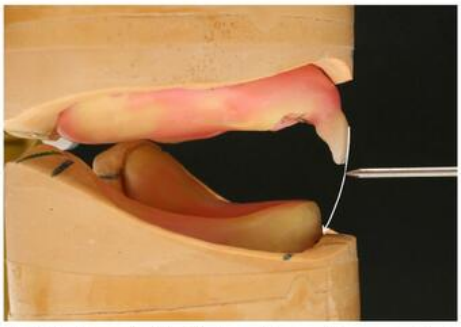


Fig 6-79 The sagittal inclination of the maxillary central incisors is set so that the incisal edges are directed towards the mandibular labial vestibule.

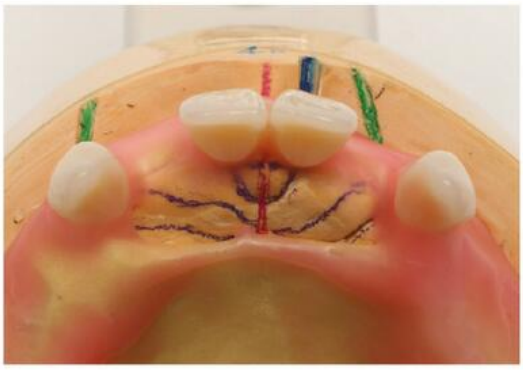


Fig 6-80 When setting the maxillary canines, their cervical areas are positioned in line with the first large pair of transverse palatine rugae. When the rugae are asymmetrical, the one on the less-resorbed side is used as reference. The distal labial surface of the canine is directed towards the posterior ridge. In patients with severe ridge resorption, the canines should be positioned 1 to 2 mm more buccally. When the rugae are hardly discernible, the set-up sequence may be altered to set the central incisors first, followed by the lateral incisors, and finally the canines.folds. The incisal edges are thus placed at 22 mm.





Fig 6-81a,b Place a 2D template on the lower member of the Stratos articulator with the curved surface facing up.



Fig 6-82 Check the horizontal alignment and symmetry of the maxillary anterior teeth using the 2D template.



Fig 6-83 Set the mandibular canines at the intersection of the reference line for the anterior ridge and that for posterior tooth set-up. Adjust the position of the canine so that its labial prominence is in line with the anterior arch mesially and with the posterior arch distally.



Fig 6-84 In the BPS denture fabrication system, the canines are set first in the mandibular arch. The cusp tip of the mandibular canine is positioned near the interproximal contact between the maxillary canine and lateral incisor on each side. This creates a smooth transition from the anterior arch to the posterior arch to facilitate the setting of the maxillary and mandibular posterior teeth and establishment of occlusal relationship.

[2] Setting the mandibular posterior teeth (set-up sequence: 14,24 → 15,25 → 16,26 →17,27)

Two lines of SR Phonares teeth are available, one for a normal bite situation with a one-tooth-to-two teeth relationship and the other for the one-tooth-to-one-tooth lingualized occlusion (**Figs 6-85,86**). In this section, tooth set-up for lingualized occlusion using SR Phonares Lingual NHC is illustrated step by step.

The template used here was developed for setting prosthetic teeth along the curve of Spee (in the sagittal plane) and the curve of Wilson (in the transverse plane). Two-dimensional (2D) and three-dimensional (3D) templates are available. When the casts are mounted on the Stratos articulator using the UTS Transfer Bow (face bow), a 3D template is used to set the posterior teeth. A 2D template is used to mount the casts without the face bow.

There are two objectives in setting the mandibular posterior teeth: to facilitate the formation of the BTC Point and to establish balanced occlusion. For these objectives, the lingual cusps of the four posterior teeth must touch the anteroposterior curvature of the 3D template. However, the buccal cusps of the mandibular posterior teeth, except the first premolar, are kept out of contact with the template. The buccal cusp of the first premolar only may be brought into contact with the template (optional for esthetics). It is essential to ensure that the buccal and lingual cusp tips are set at the same height as the occlusal plane (**Figs 6-87~94**).

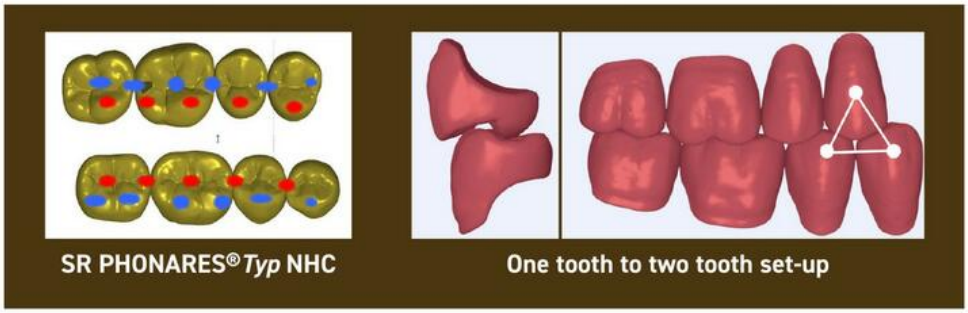


Fig 6-85 One-tooth-to-two-teeth (cusp-to-ridge) occlusion.

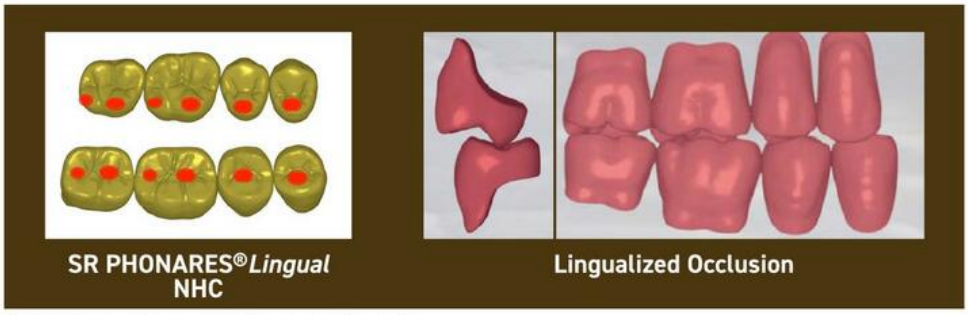


Fig 6-86 One-tooth-to-one-tooth (cusp-to-fossa) occlusion.

Laboratory procedure



Fig 6-87 Attach a 2D or 3D template to the upper member of the Stratos articulator. The 3D template, in particular, must be secured by tightening a fixation screw with a screwdriver (photo shows 3D template).

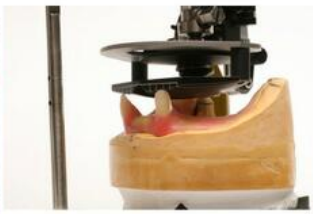


Fig 6-88 Set the 3D template at distal 1/3 of the retromolar pads and the cusp tips of the canines.



Fig 6-89 The buccal cusp of the first premolar only is brought into contact with the template. The buccal cusps of the other posterior teeth do not touch the template.



Fig 6-90 The tooth axes of all four posterior teeth are set perpendicular to the template. There should be a clearance between the template and the buccal cusp tips of the posterior teeth except for the first premolar. Set the buccal and lingual cusps on the same plane at the height of the occlusal plane.



Fig 6-91 Align the central grooves of the posterior teeth with the centerline of the alveolar ridge.



Fig 6-92 Pound's line is a line connecting the lingual of the retromolar pad and the mesial of the canine. Denture teeth should never be set lingual to Pound's line.



Fig 6-93 Frontal view of the posterior teeth arranged with the 3D template.



Fig 6-94 Occlusal view of the completed set-up of the posterior teeth.

[3] Setting the maxillary posterior teeth (set-up sequence:16,26 → 14,24 → 15,25 →17,27) (Fig 6-95)

When setting the maxillary posterior teeth in lingualized occlusion, they are arranged in the order of (1) the first molars, (2) the first premolars, (3) the second premolars and (4) the second molars. Intercuspation is established by starting with the first molars, followed by the first premolars that might affect facial esthetics, then the second molars, and finally, the second molars.

6: The mesial palatal cusp of the maxillary first molar is placed in contact with the central fossa of the mandibular first molar and the distal palatal cusp with the distal fossa of the mandibular first molar.

4: The palatal cusp of the maxillary first premolar engages with the central fossa of the mandibular first premolar. From an esthetic standpoint, the inner incline of the maxillary buccal cusp may be brought into contact with the outer incline of the mandibular buccal cusp.

5: The palatal cusp of the maxillary second premolar is positioned in contact with the central fossa of the mandibular second premolar.

7: The palatal cusp of the maxillary second molar engages with the central fossa of the mandibular second molar. The buccal cusps are positioned to follow the curve of Monson.

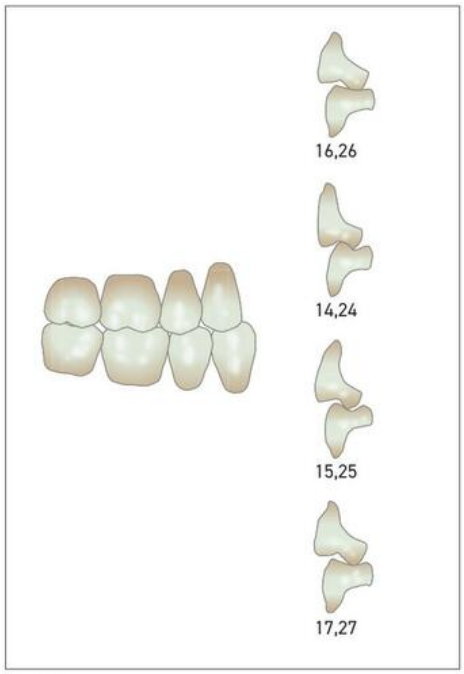


Fig 6-95

Laboratory procedure





Fig 6-96a,b The first molars are placed in a one-tooth-to-one-tooth relationship.

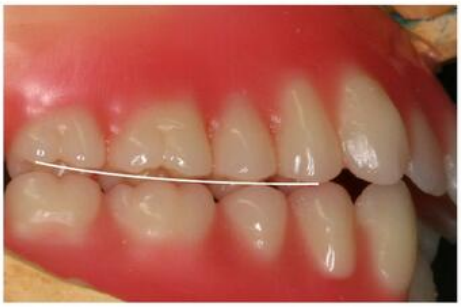



Fig 6-97a,b The completed set-up of the maxillary posterior teeth. The buccal cusps follow the Monson curve.

Setting the mandibular incisors

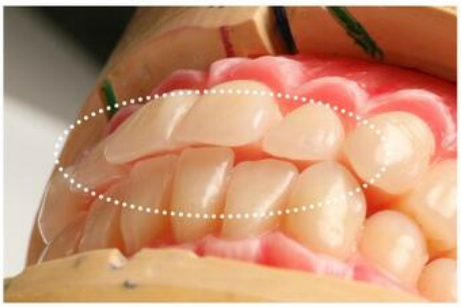


Fig 6-98 Highly individualized set-up of the mandibular incisors with a reduced clearance between the maxillary and mandibular incisors.

Completed set-up of all teeth




Fig 6-99a,b Cusp-to-fossa relationship of the maxillary teeth to the mandibular teeth in lingualized occlusion.





Fig 6-100a–c The completed set-up of the denture teeth in lingualized occlusion.

[4] Occlusal check after set-up

Laboratory procedure

Checking the centric contacts



Fig 6-101a~c Five occlusal contacts on the posterior teeth per side (one additional contact on the buccal cusp of the first premolar is optional).

Occlusal adjustments for protrusive and lateral movements

Protrusive movement



Fig 6-102a~c Adjustments in protrusive movement: Adjustments are made so that anterior and posterior teeth contact and glide at the same time.

Lateral movements



Fig 6-103a~c Adjustments in left lateral movement. Bilateral balanced occlusion, whereby both the right and left posterior teeth come into contact and glide simultaneously, is provided. Adjustments are made in right lateral movement in the same manner.

14 3 shape digital scanning system

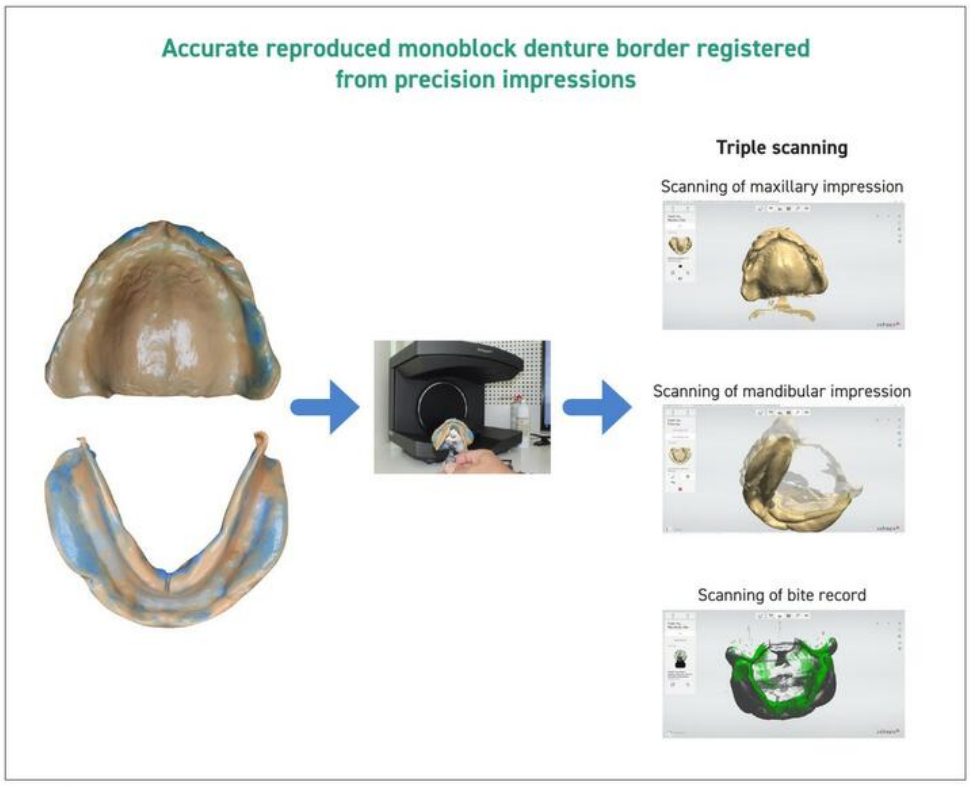


Fig 6-104

15 Digital tooth Set-up

[1] Tooth setup by the digital denture software (Shape E3 and Ivoclar Vivadent Digital Denture Professional Add-on)

Comparing digital denture production and conventional analogue method, it is certain that total production time for setting up denture teeth can be dramatically reduced due to its algorithm.

At the initial design process the amount of block out required and the tooth positioning are digitally set. Following the fundamental setup, tooth positioning is automatically determined based on a predetermined occlusal plane and anatomical landmarks. In addition to using a preformed denture teeth setup from the software, it is possible to modify individual teeth or as a whole arch by following the conventional method such as pound technique. For conducting an alternation of the arch form or each digital denture tooth, it is recommended to apply conventional knowledge to create targeted occlusion. Furthermore in order to maximize the potential of the digital denture production, it is critical to duplicate an effective shape of polishing surface to maximize the suction effectiveness of the mandibular complete denture.

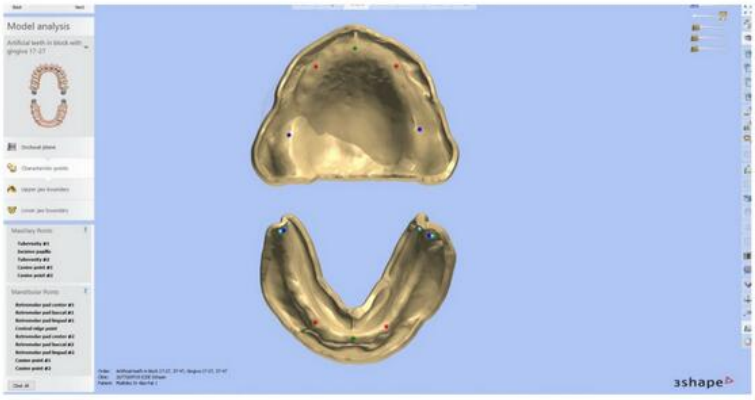


Fig 6-105 Anatomical landmarks are pointed out.

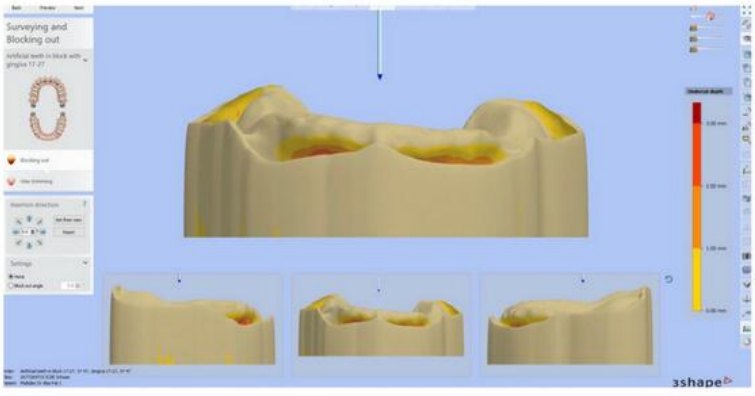


Fig 6-106 Denture insertion path is determined and undercuts are blocked out.

In order to determine ideal teeth positioning, first, the path of insertion was decided digitally. It is determined by analyzing how much undercut is to be blocked out by altering the angle of the model digitally. Next denture teeth are selected from a tooth library to customize individuality. Then an alternation of occlusal positioning is conducted. e.g. Digitally modifying occlusion to be cusp to marginal ridge (one tooth-to-two teeth), cusp to fossa (one tooth-to-one tooth) or removal of premature contact. In general, ideal teeth positioning is automatically determined by analyzing tentative occlusal plane and anatomical landmarks. These tooth positions if necessary, can be selectively modified/ altered as an arch or as an individual tooth. To modify these positions, it is advisable to apply conventional wisdom (analogue denture knowledge) to enhance the quality of occlusion and individuality. By applying your analogue skills into digital denture production, it enhances the overall quality of result.

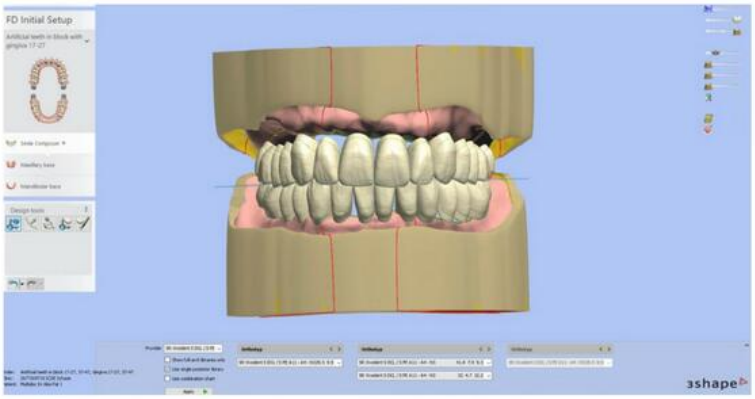


Fig 6-107 The selected artificial teeth were aligned on the residual ridge.

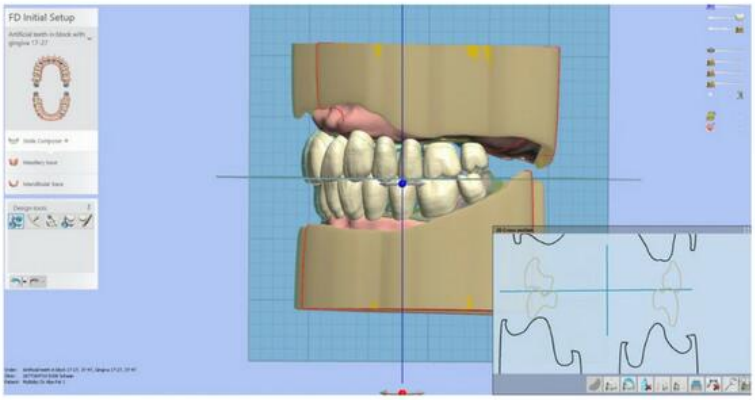


Fig 6-108 The artificial teeth can be arranged to the adequate position while observing the cross section for the top and bottom alveolar crest.

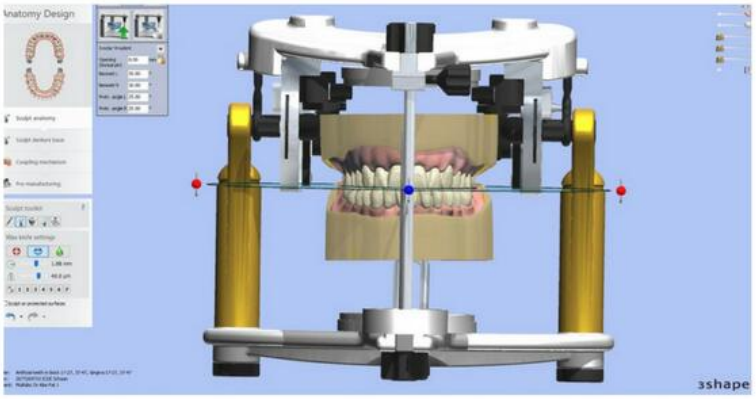


Fig 6-109 By the movement of the average articulator, cusp interference can be found and removed.

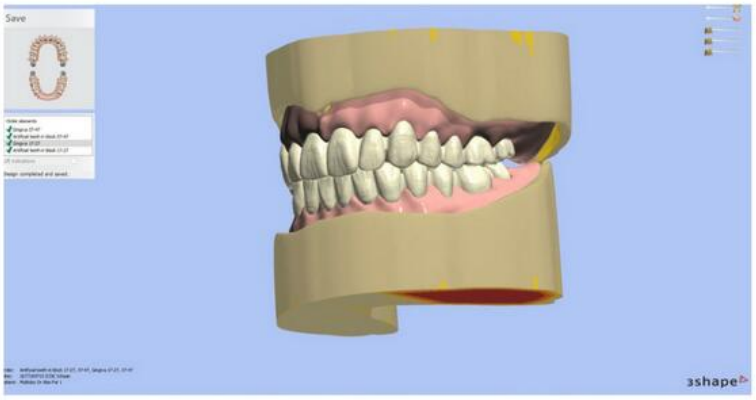


Fig 6-110 Gingiva-polished surface can be provided the way you want in order to make the natural contact of mucosal tissue and denture labial, buccal and lingual polished surfaces.